

LESSON 16 / 第十六課

Instrument Monitoring of Trees: From Networks to Alarm Thresholds

樹木的儀器監測：從網絡到警報閾值

Format. Short literature review. Claims that carry a number or a causal ranking are tied to a numbered source. Scale runs from the largest published pattern to the finest measured process. The lesson is not sequenced from the previous number and is not a regional case file unless the object itself is regional.

體例。短文獻綜述。帶數字或因果排序的陳述繫到編號文獻。尺度由最大已發表格局寫到最細已測過程。本課不由上一號遞推；除非對象本身是地域性的，否則不是單一地區檔案。

Abstract

摘要

Sensors produce time series, not capacity. Static tilt is not gust-by-gust rotation. Alarms need declared error rates. Multi-year, multi-species validation against actual failure remains short. Coverage is a sampling design; unrepresentative sites plot the wrong population. Baselines are species- and site-specific. A dashboard is a record. It is not a verdict and not new section. Integration belongs inside the same residual-risk sentence used for every other tool.

傳感器產生時間序列，不是容量。靜力傾斜不是一陣一陣轉動。警報需要宣布誤差率。多年多樹種對實際失效的驗證仍短。覆蓋是抽樣設計；無代表性立地畫出錯誤族群。基線因樹種與立地而異。儀表板是紀錄。它不是判決，也不是新截面。整合屬於與其他工具相同的剩餘風險句子之內。

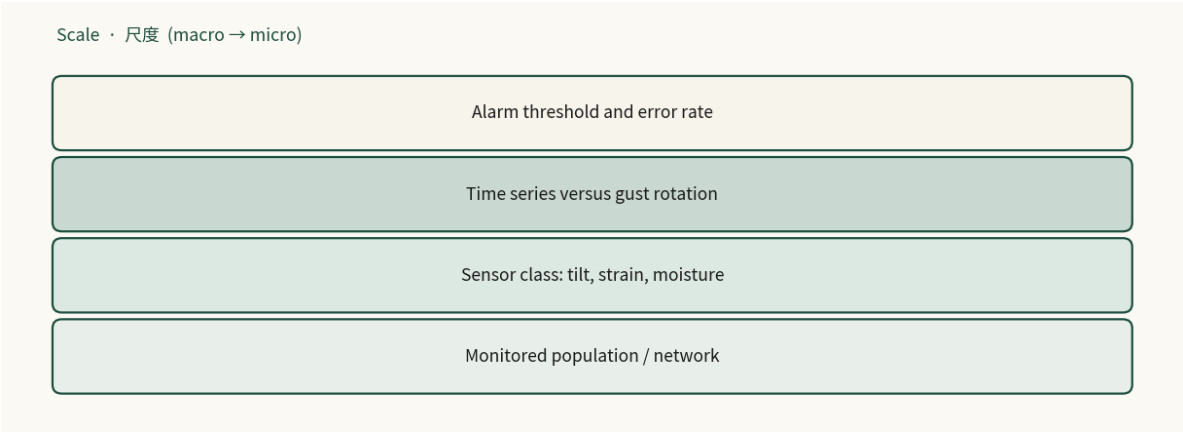


Figure 1. Nested scales from the monitored network to the alarm.

圖 1. 由監測網絡到警報的嵌套尺度。

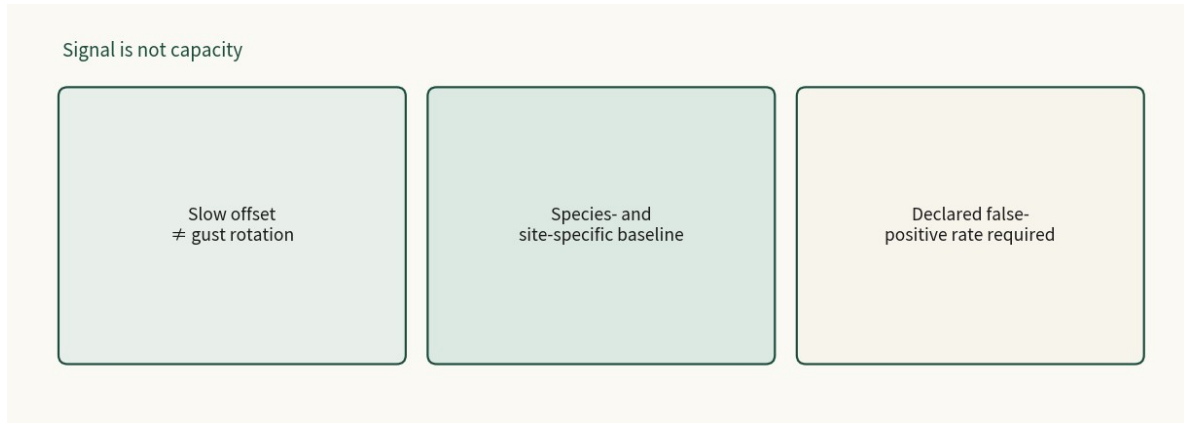


Figure 2. Offset, baseline and error rate are different objects.

圖2. 偏移、基線與誤差率是不同對象。

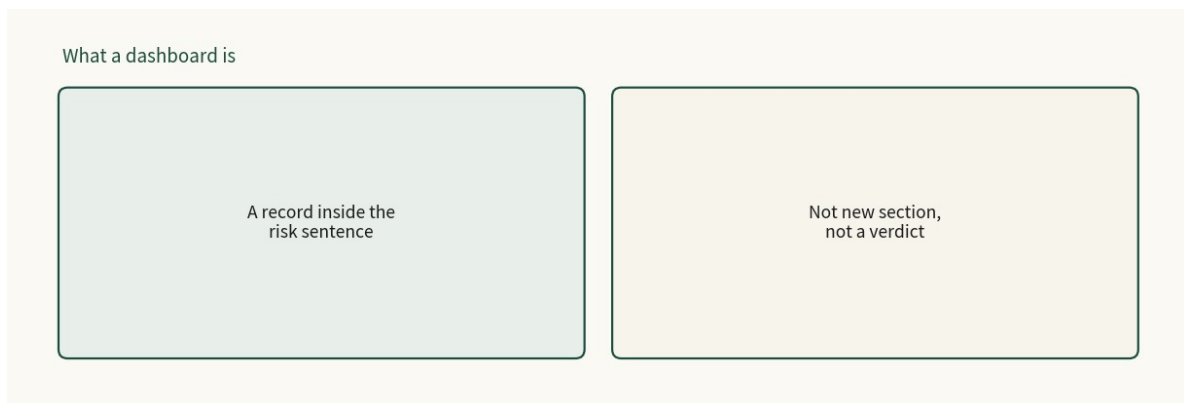


Figure 3. A dashboard is a record, not a verdict.

圖3. 儀表板是紀錄，不是判決。

1. Scope

1. 範圍

The object is published sensor use on standing trees: networks, signals, thresholds. A municipal pilot is an example of coverage, not a validation against failure.

對象是站立樹上傳感器的已發表使用：網絡、信號、閾值。市政試點是覆蓋的例子，不是對失效的驗證。

2. Sampling and signal

2. 抽樣與信號

Coverage is a sampling design [1]. Unrepresentative sites plot the wrong population. Slow offset is not gust-by-gust rotation [2]. Baselines are species- and site-specific [3]. Treating a one-week tilt as a capacity rewrite confuses a record with a section modulus.

覆蓋是抽樣設計 [1]。無代表性立地畫出錯誤族群。慢偏移不是一陣一陣轉動 [2]。基線因樹種與立地而異 [3]。把一週傾斜當成容量改寫，是把紀錄與截面模量混淆。

3. Thresholds and the sentence

3. 閾值與句子

No declared false-positive rate means no method. Sensors do not see plate saturation directly. Integration belongs inside the Level-2 / Level-3 residual-risk sentence: the series informs likelihood of a specified part in a named envelope. It does not abolish targets.

沒有宣布假陽性率即沒有方法。傳感器不直接看見盤飽和。整合屬於第 2／第 3 級剩餘風險句子之內：序列為點名包絡中指定部位的可能性提供資訊。它不取消目標。

4. Limits

4. 限度

Multi-year, multi-species validation against actual failure remains short. A pilot of thousands of trees that has not yet been followed to failure is a coverage claim, not a calibration curve.

多年多樹種對實際失效的驗證仍短。尚未追隨到失效的數千株試點，是覆蓋主張，不是率定曲線。

Study questions

研習題

Choose the option that retains the published result. Answers follow the last item.

選出保留已發表結果的選項。答案在最後一題之後。

1. A tilt series is:

1. 傾斜序列是：

A. New capacity.

A. 新容量。

B. A record.

B. 一項紀錄。

C. Wall 4.

C. 第 4 壁。

D. P88.

D. P88。

2. Static vs gust:

2. 靜力對陣風：

A. Same object.

A. 同一物件。

B. Different objects.

B. 不同物件。

C. Both P50.

C. 都是 P50。

D. 3-30-300.

D. 3-30-300。

3. Alarms need:

3. 警報需要：

A. No error rate.

A. 無誤差率。

B. A declared error rate.

B. 宣布的誤差率。

C. Only paint.

C. 只有漆。

D. Invoices.

D. 發票。

4. Saturation is seen:

4. 飽和被看見：

A. Directly always.

A. 總直接。

B. Not directly.

B. 不直接。

C. As Wall 1.

C. 作為第 1 壁。

D. As flash density.

D. 作為閃擊密度。

5. Network design is:

5. 網絡設計是：

A. Optional.

A. 可選。

B. Sampling.

B. 抽樣。

C. A flush cut.

C. 平切。

D. t/R.

D. t/R。

6. Baselines should be:

6. 基線應當：

A. Universal.

A. 萬能。

B. Species- and site-aware.

B. 對樹種與立地有意識。

C. Always zero.

C. 總是零。

D. 10-20-30.

D. 10-20-30。

7. Failure validation is:

7. 失效驗證：

A. Complete.

A. 完整。

B. Still short.

B. 仍短。

C. Illegal.

- C. 非法。
- D. A cable grade.
- D. 鋼纜等級。

8. A dashboard is:

8. 儀表板是：

- A. A verdict.
- A. 判決。
- B. Not a verdict.
- B. 不是判決。
- C. A target.
- C. 一個目標。
- D. SOS.
- D. SOS。

9. A pilot not followed to failure is:

9. 尚未追隨到失效的試點是：

- A. A calibration curve.
- A. 率定曲線。
- B. A coverage claim.
- B. 覆蓋主張。
- C. Wall 3.
- C. 第 3 壁。
- D. A flare datum.
- D. 幹基基準。

10. Sensors abolish targets:

10. 傳感器取消目標：

- A. Yes.
- A. 是。
- B. No.
- B. 否。
- C. Only in palms.
- C. 只在棕櫚。
- D. If wireless.
- D. 若無線。

Answers / 答案 1-B 2-B 3-B 4-B 5-B 6-B 7-B 8-B 9-B 10-B

References

參考文獻

- [1] Municipal tilt-monitoring pilots and sampling-design notes.
- [2] James K.R. inclinometry / sway papers.
- [3] Reviews of IoT and strain monitoring on standing trees.
- [4] ANSI A300 Part 9 — instruments as Level-3 inputs, not verdicts.

Literature review · Lesson 16 · 文獻綜述 · Macro → micro

Closing note. This lesson is written as a short review article. Disagreement in the literature is kept in view where the results actually diverge.

結語。本課以短綜述文章寫成。文獻真正分歧之處，兩造結果都保留在視野內。